K Nearest Neighbour Classification

Query => X = (Maths = 6, CS = 8), K=3

	Maths	CS	Result
1)	4	3	Fail
2)	6	7	Pass
3)	7	8	Pass
4)	5	5	Fail
5)	8	8	Pass

Euclidean Distance:

$$d = \sqrt{|x_{o1} - x_{A1}|^2 + |x_{02} - x_{A2}|^2}$$

I) $\sqrt{(6-4)^2 + (8-3)^2} = \sqrt{29} = 5.38$

II) $\sqrt{(6-6)^2 + (8-7)^2} = 1$

III) $\sqrt{(6-7)^2 + (8-8)^2} = 1$

IV) $\sqrt{(6-5)^2 + (8-5)^2} = \sqrt{10} = 3.16$

V) $\sqrt{(6-8)^2 + (8-8)^2} = 2$

P P P = 3 Pass
0 Fail
Since 3>0 X is considered as Pass